

Revised Generation Interconnection System Impact Study Report

For Queue Project Q11-0023

Elkview Coal 230 kV
November 22, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0023. The Interconnection Customer proposes a 242.3 MW wind facility interconnecting at the Elkview Coal 230 kV substation in the Ohio area. The request was received on March 30, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 18 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$9,347,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: OLYMPUS 138 kV/1 kV TX
Contingency: INTERNATIONAL PAPER – FLATLICK 500kV
Loading: 152.8% (Rating: 142 MVA)
DFAX: 7.3%
Pre-project loading: 140.5%
2. Facility: SPORN – GAVIN POWER LLC SUBSTATION 230kV
Contingency: INTERNATIONAL PAPER – FLATLICK 500kV
Loading: 103.5% (Rating: 520 MVA)
DFAX: 36.7%
Pre-project loading: 86.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
OLYMPUS 138/1 kV Transformer	Replace transformer with higher-rated unit	\$5,017,000	100% (\$5,017,000)
SPORN – GAVIN POWER LLC 230kV	Reconductor line	\$4,330,000	100% (\$4,330,000)
	Total allocated to project		\$9,347,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.